HACKATHON DEMO GUIDE

FairFlow Guardian

An explainable AI trading copilot that decides when to trade, when to reduce risk, and when the fairest action is no trade.

Multi-agent review | Fairness Passport | AI committee | Verifiable audit hash

The one-line explanation

FairFlow is not a pure profit bot. It is a safety, fairness, and transparency layer for retail traders: it proposes a trade only after market parity, manipulation risk, stress tests, and execution guardrails all pass.

Current demo behavior

CALM SCENARIO	MANIPULATED SCENARIO	PAPER ORDER
APPROVED	NO_TRADE	GATED

The demo is designed to show judgment, not bravado: a calm setup can be approved, while unsafe conditions are blocked with reasons.

BGA AI TRADING AND STRATEGY TRACK

What problem is it solving?

Retail traders often see the same price chart as institutions, but not the same risk context.

Typical trading bot

- Chases signals and headline returns.
- Often hides assumptions and overfitting.
- Can keep trading in thin or manipulated markets.

FairFlow Guardian

- Treats no trade as a valid answer.
- Explains every approval or rejection.
- Caps execution with risk and liquidity safeguards.

Why this fits BGA

- Reduces information asymmetry.
- Makes strategy decisions inspectable.
- Optimizes for healthier market behavior.

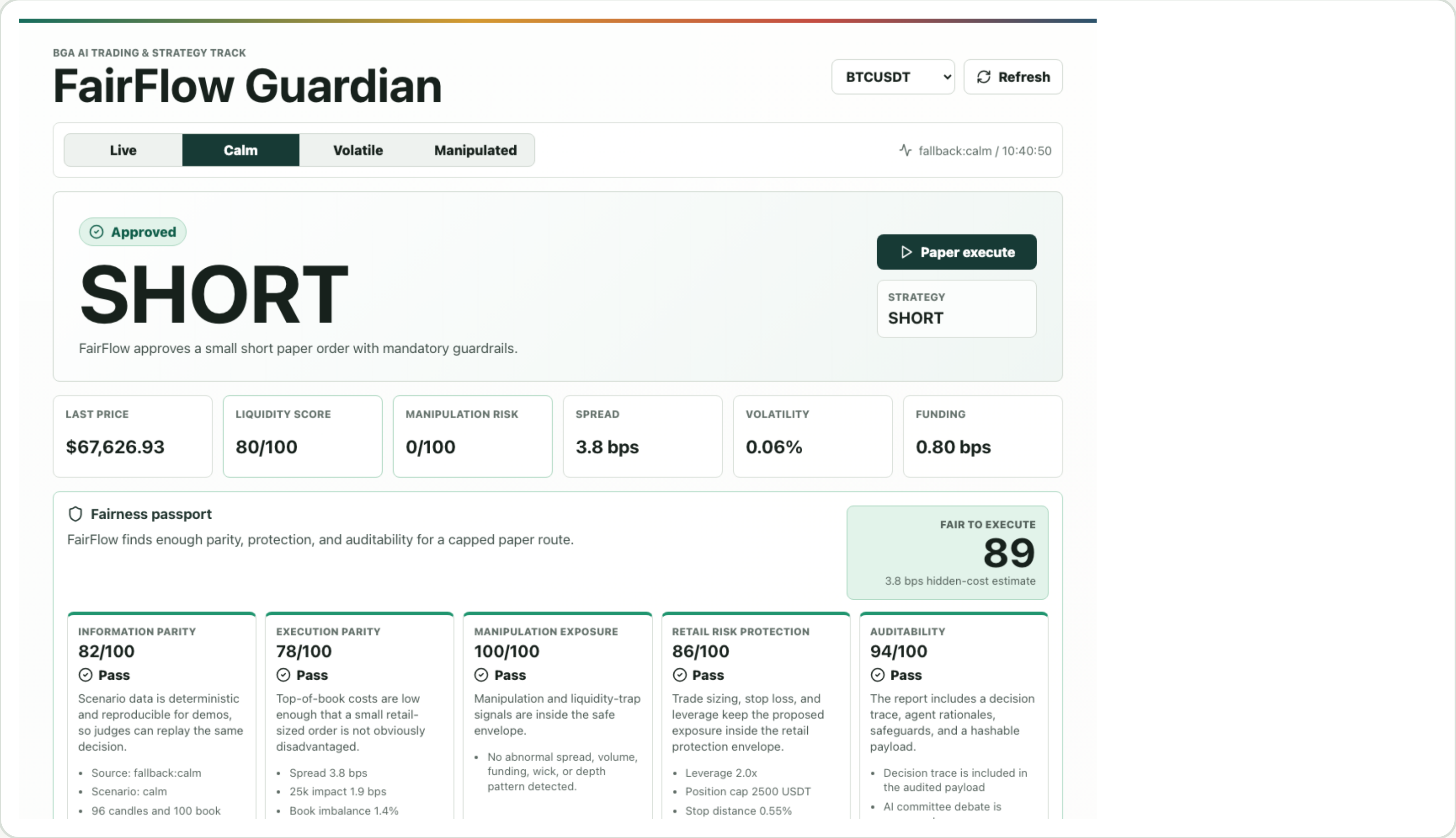
The project asks a better question than: How do we maximize PnL? It asks: Is this trade fair, explainable, and safe enough to execute?

- For a judge: this is a market safety system, not a black-box alpha claim.
- For a user: this is a trading seatbelt that explains risk before money moves.
- For a protocol: this creates a verifiable record of why execution did or did not happen.

BGA AI TRADING AND STRATEGY TRACK

What you are seeing in the UI

The dashboard is a decision cockpit. Each button runs the same review process on a different market condition.



1. Scenario selector

Live uses Bybit public data. Calm, Volatile, and Manipulated are deterministic demos.

2. Final action

Approved can create a paper route. Blocked or Observe locks execution.

3. Fairness passport

Scores retail parity, hidden costs, manipulation exposure, and auditability.

4. Decision trace

Seven stages show how raw data becomes a final audited outcome.

5. Audit hash

The full report is hashed so the team cannot rewrite it after the outcome is known.

How the system works

FairFlow separates proposal from approval. The strategy agent cannot execute unless independent safety agents agree.

Bybit public API or fallback scenario → **Market metrics** → **Deterministic agents** → **Agentic AI committee** → **Fairness passport and gate** → **Report and SHA-256 hash**

Market Regime Agent

Classifies whether the market is orderly, directional, volatile, or fragile.

Strategy Agent

Suggests long, short, or no trade with entry, stop, take-profit, and confidence.

Manipulation Sentinel

Scores abnormal spread, depth imbalance, wick, volume, impact, and funding crowding.

Risk Guardian

Stress-tests the proposal and blocks trades that violate safety thresholds.

AI Committee

Adds ML regime, anomaly, forecast, backtest, planner, memory, and narration.

Decision Trace

Records the seven-stage path from market data intake to final audited outcome.

What each agent is checking

The project is intentionally inspectable: every score comes from visible market conditions and explicit thresholds.

1 Market Regime

- Volatility
- 24h range
- Momentum
- RSI

2 Strategy

- Action
- Entry and stop
- Take profit
- Confidence

3 Manipulation Sentinel

- Spread
- Book imbalance
- Wick and volume
- Crowded funding

4 Risk Guardian

- Stress moves
- Risk budget
- Position cap
- Execution lock

5 Fairness Passport

- Information parity
- Execution parity
- Manipulation exposure
- Auditability

6 Execution and Audit

- Planner
- Memory
- Narrator
- Decision hash

How to demo it in two minutes

This is the exact sequence to walk a judge through the app without getting lost in trading jargon.

1. **Open BTCUSDT in Calm**
   Show that the system approves a small paper trade only after all safety agents pass.

2. **Click Paper execute**
   Point out that this is gated paper execution, not uncontrolled live trading.

3. **Open Fairness Passport**
   Show the retail parity score and the hidden-cost estimate before discussing the trade.

4. **Open Decision trace**
   Walk through data intake, features, proposal, AI committee, stress decision, and final outcome.

5. **Switch to Manipulated**
   Show that the same strategy idea is rejected when spread, funding, wick, and depth become unsafe.

Talk track

The important behavior is not that FairFlow can trade. It is that FairFlow knows when not to trade. The system separates alpha generation from fairness review and risk approval, shows a seven-stage decision trace, and creates an audit trail so the decision can be verified later.

Example audit prefix: 97376071cee440c2...

What the scenarios prove

The same app can approve, observe, or block depending on market quality and risk context.

SCENARIO	DECISION	LIQUIDITY	FAIRNESS	VOLATILITY	WHY IT MATTERS
Calm	APPROVED / SHORT	80/100	89/100	0.06%	FairFlow approves a small short paper order with mandatory guardrails.
Volatile	BLOCKED / NO_TRADE	51/100	67/100	0.81%	FairFlow blocks execution after AI committee review: ML anomaly detector flagged extreme market-structure risk.
Manipulated	BLOCKED / NO_TRADE	28/100	50/100	2.04%	FairFlow blocks execution after AI committee review: ML regime classifier detected fragile liquidity.

Judge-friendly interpretation: FairFlow demonstrates selective execution. It does not pretend every signal deserves a trade.

Risk management is the product

The execution layer is deliberately conservative because crypto leverage can fail fast.

Built-in safeguards

- Paper/testnet execution only in this prototype.
- Stop loss is mandatory for any executable recommendation.
- Position size is capped at 25% of demo equity.
- Maximum modeled risk budget is 1% of demo equity.
- Execution is blocked during high manipulation or fragile-liquidity states.
- Cooldown-first design: one decision report per user refresh, no runaway loop.

Stress test example

MOVE	PNL	STOP
-1%	+2.0%	No
-2%	+4.0%	No
+1%	-2.0%	Yes
+2%	-4.0%	Yes
+5%	-10.0%	Yes

For the calm short, adverse upward moves trigger the modeled stop. In manipulated conditions, execution is blocked before stress tests matter.

Transparency and verifiability

A good market system should be inspectable before the outcome is known.

Decision report → **Canonical JSON** → **SHA-256 hash** → **Optional contract anchor**

What is inside the report?

- Market snapshot source
- Metric values
- Fairness Passport
- Agent verdicts
- Decision trace stages
- Stress tests

What does the contract add?

- Stores the decision hash.
- Records symbol, action, reporter, timestamp, and metadata URI.
- Makes hindsight rewriting much harder.

How this maps to judging criteria

Use this page as the quick mental checklist before presenting.

BGA ethos	Uses a Fairness Passport to help retail users avoid unsafe trades, reduce hidden information gaps, and reward transparent no-trade decisions.
Technical depth	FastAPI agents, AI committee models, Bybit V5 market data, deterministic fallback scenarios, React dashboard, and Solidity audit anchoring.
Strategy and risk	Strategy proposals are separate from approval, with fairness scoring, stop loss, stress tests, liquidity thresholds, and manipulation gates.
Transparency	Every recommendation has a decision trace, agent rationales, metric values, safeguards, and a SHA-256 audit hash.

Simple pitch script

A concise explanation you can read almost word for word during the demo.

FairFlow Guardian is an explainable AI trading safety layer for retail crypto traders. Most bots focus on whether they can make money. We focus on whether a trade is safe, fair, and explainable enough to execute. The system pulls market data, builds metrics like spread, volatility, funding, depth, RSI, and market impact, then sends the setup through deterministic agents and an AI committee. One agent proposes a strategy, but separate agents challenge it for regime risk, manipulation risk, forecast uncertainty, backtest weakness, and stress-test failure. If the setup is unfair or unsafe, FairFlow blocks execution and records the full decision trace. If it passes, it creates only a capped paper order with stop loss and an audit hash.

Is this a profit bot?

No. It is a risk and transparency layer. Profit models can plug in later, but execution remains gated.

Why blockchain?

The report hash can be anchored on-chain, proving the decision existed before the outcome.

Why is it useful?

It gives retail users institutional-style context: market quality, liquidation pressure, and explainable reasons to wait.